Experiment 5: Average Fuel Efficiency and Percentage Improvement

Aim

To calculate the average fuel efficiency of car models and determine the percentage improvement in fuel efficiency between two car models using NumPy.

Algorithm

1. **Start the program.**
2. **Import the NumPy library.**
3. **Create a NumPy array containing the fuel efficiency of different car models.**
4. **Calculate the average fuel efficiency using np.mean().**
5. **Calculate the percentage improvement between two car models.**
6. **Display the average fuel efficiency and percentage improvement.**
7. **Stop the program.**

Code

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35])

average_efficiency = np.mean(fuel_efficiency)

percentage_improvement = (
   (fuel_efficiency[1] - fuel_efficiency[0])
   / fuel_efficiency[0]
) * 100

print("Average fuel efficiency:",
    f"{average_efficiency:.2f} MPG")

print("Percentage improvement from Model 1 to Model 2:",
    f"{percentage_improvement:.2f}%")
```

Input

Model 1 = 25 MPG

Model 2 = 30 MPG

Model 3 = 35 MPG

Output

Average fuel efficiency: 30.00 MPG

Percentage improvement from Model 1 to Model 2: 20.00%

```
====== RESTART: C:/Users/sivak/Desktop/FODS LAB experiments codes/exp 5.py =====
Average fuel efficiency: 30.00 MPG
Percentage improvement from Model 1 to Model 2: 20.00%
>
```

Result

Thus, the average fuel efficiency was calculated as 30 MPG, and Model 2 showed a 20% improvement over Model 1.